



Sensing-Assisted Beam Tracking and Secret Key Generation

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Outline of presentation

I. Introduction

II. System model

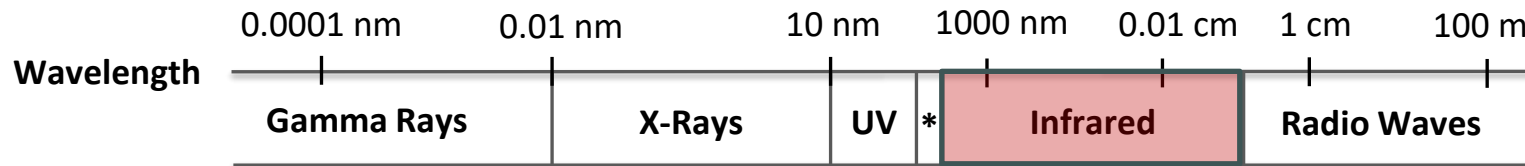
III. Simulation results and Conclusion

Free-Space Optics (FSO): Advantages and Challenges

○ For high-speed connections, FSO is a promising candidate

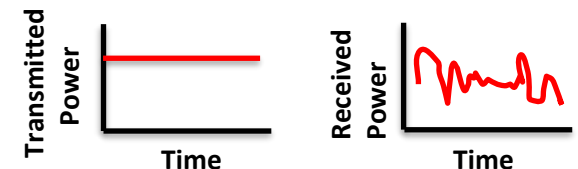
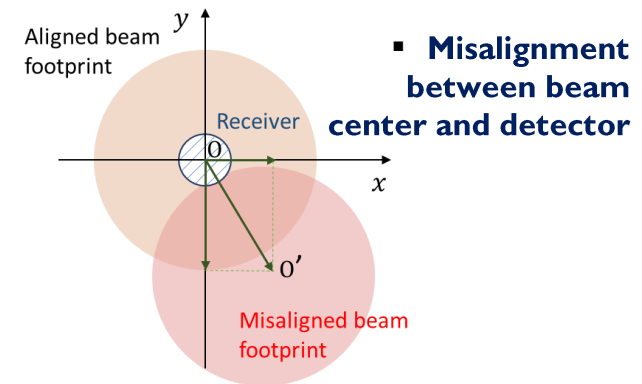
- Using infrared bands → support extremely high data rates ($\sim$ Gbps, up to Tbps)
- Low power consumption and License-free spectrum

(*): Visible light



○ Challenges in implementation of FSO (I)

- Atmospheric turbulence, Atmospheric attenuation
- Pointing errors: Cause a decrease and fluctuation in the received signal
→ Degrade the performance of FSO systems
- Pointing errors are caused by:
 - + Mechanical vibrations of transceiver platforms (e.g., UAVs, satellites)
 - + Imperfect beam tracking and alignment systems
 - + Atmospheric turbulence causing beam wander



➡ What is the solutions for pointing errors?

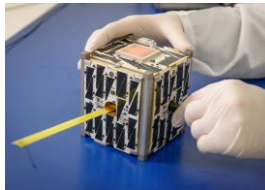
Pointing errors solution — overview of APT mechanisms

○ Definition of APT mechanisms:

- An Acquisition, Pointing, and Tracking (APT) mechanism is a system used in FSO communications to establish, align, and maintain a reliable line-of-sight optical link between transceivers

○ Limitation:

- Traditional APT systems often suffer from high complexity, power consumption, and payload requirements [2]
→ Making them less suitable for lightweight UAVs and small satellite (CubeSats)



→ A CubeSat built at NASA's Ames Research Center measures about 10 centimeters in a side.

○ Alternative solution: Sensing-assisted optical wireless communication

- **Key idea:** Time-division multiplexing is used to allow sensing and the communication to share the same transceiver hardware, where sensing phase acts as the function of an APT system.



Using same hardware

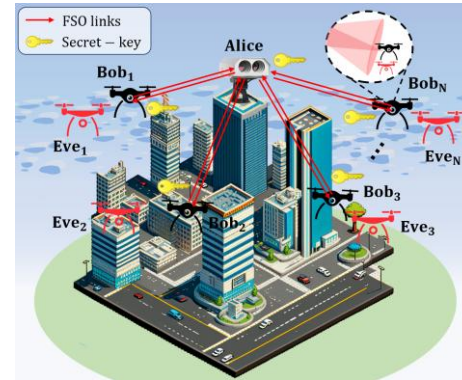
→ Reduce resource consumption → Suitable for lightweight UAVs and small satellites

Free-Space Optics (FSO): Advantages and Challenges

Security Challenges in FSO (II)

- FSO operates over an open and broadcast medium
→ Optical signals can be intercepted by eavesdropper

➔ Key question: How can we address security challenges in FSO?
Secure key establishment is one possible solution

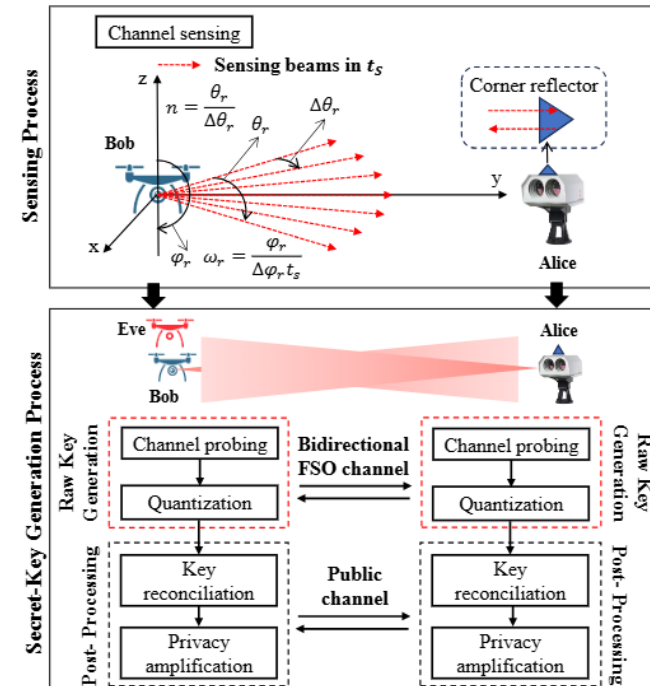


APWCS conference version

- Objective:** Develop a sensing-assisted FSO framework that performs beam alignment before generating secret keys from atmospheric turbulence.
- Limitation:**
 - + SKR of the system is limited by $1/Tc \rightarrow$ Low SKR; (where Tc : channel coherent time)
 - + No beam tracking after sensing

APWCS extension

- Objective**
 - + Increase SKR using multiple lenses
 - + Enable beam tracking after sensing by using pointing error estimation



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System model

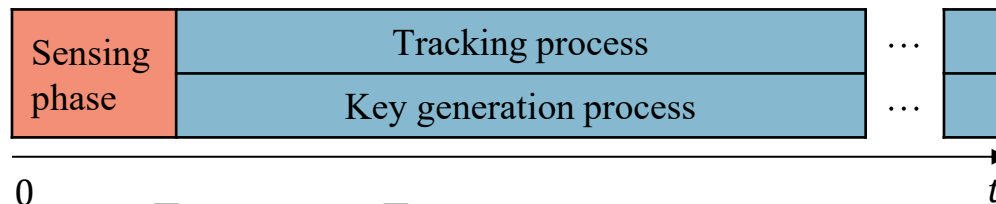
➤ System model:

- Bob: A hovering UAV without an APT system
- Alice: Base station is equipped with an APT system
- Alice and Bob generate secret keys from the estimated turbulence channel.

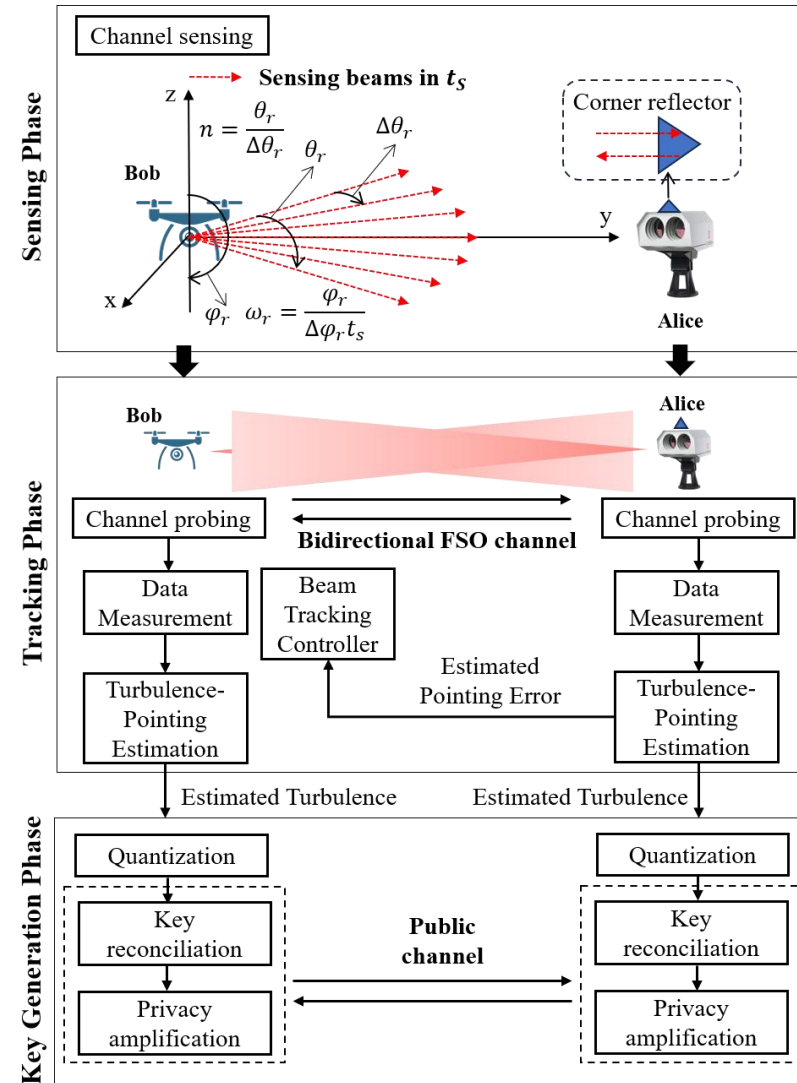
Propose scheme

1. **Sensing phase:** UAV accurately estimates the receiver's location
2. **Tracking phase:** Alice send estimated pointing error to Bob for beam tracking controller
3. **Key generation phase:** Alice and Bob extract the randomness of the estimated turbulence channel to generate secret keys

*Tracking and key generation operate in parallel



$$T_{\text{sensing}} \ll T_{\text{Key_generation_process}}$$



System model

I. Sensing phase:

- **Target:** Bob estimates azimuth and pitch angles of BS (Alice)
- **Method:** multi-line (Horizontal scanning) and rotational optical scanning (vertical scanning)

□ Horizontal scanning → azimuth angle

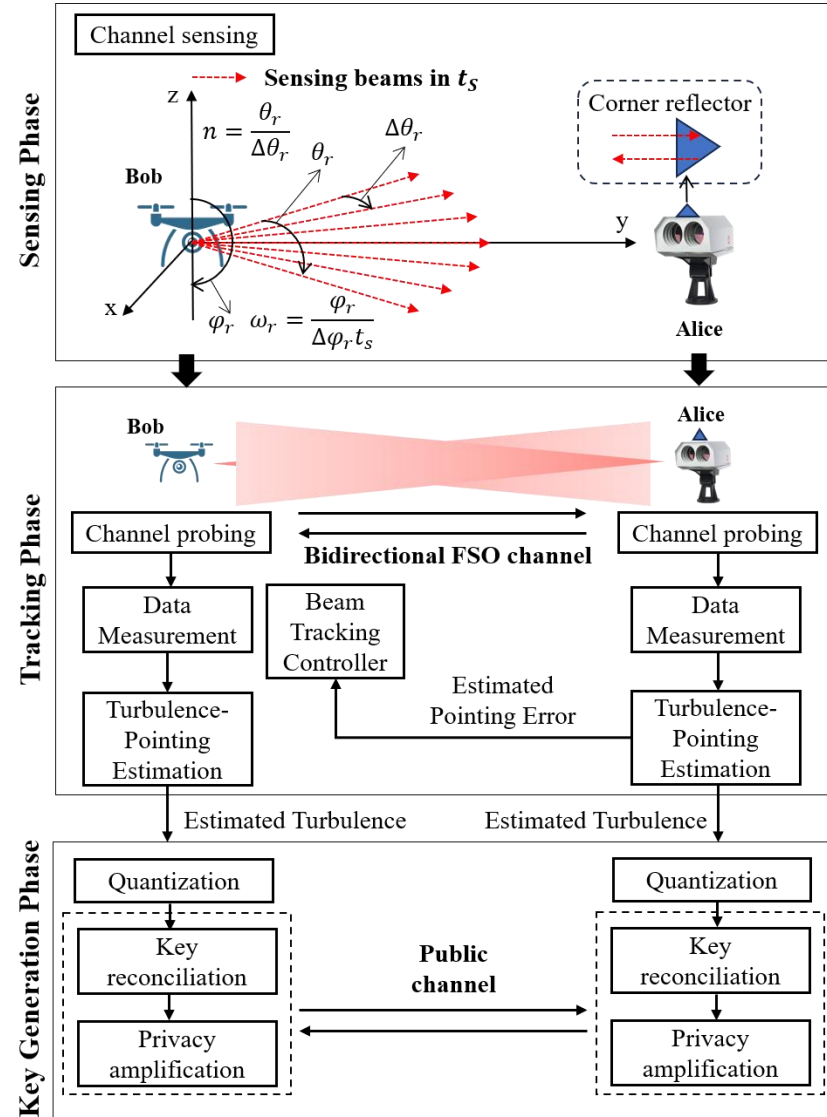
- Horizontal scanning range: θ_r
- Number of lines in the scanning: n
- Horizontal scanning step: $|\Delta \theta_r| = \theta_r/n$

□ Vertical scanning → pitch angles

- Vertical scanning range: φ_r
- Vertical scanning step: $\Delta \varphi_r = \varphi_r/(\omega_r t_s)$

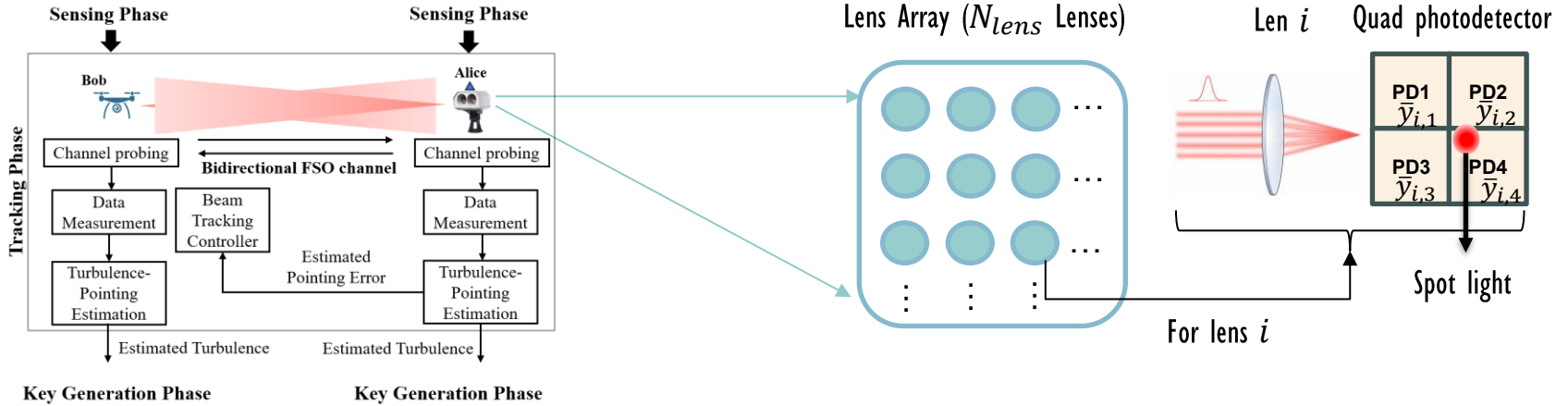
Where: the scanning frequency (ω_r) and sensing time (t_s)

→ The UAV estimates the location of the BS using reflected signals from the corner reflector.



System model

2. Tracking phase:



I) The channel between Alice and Bob are affected by atmospheric turbulence, pointing error and atmospheric attenuation

- The pointing error at Alice is caused by

+ Transmitter UAV tracking error (Bob): $\theta_e = (\theta_{ex}, \theta_{ey})$

+ Received-induced angular jitter (Alice): $\theta_r = (\theta_{rx}, \theta_{ry})$

- The instantaneous angle of arrival (AoA) at Alice:

$$\theta_{AoA} = (\theta_{AoA,x}, \theta_{AoA,y}) = \theta_e + \theta_r$$

II) Overall channel model (Bob to Alice)

- The channel at PD j behind lens i is: $y_{i,j} = P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y}) h_{a,i} + n_{i,j}$

- The pointing gain coefficient of lens i :

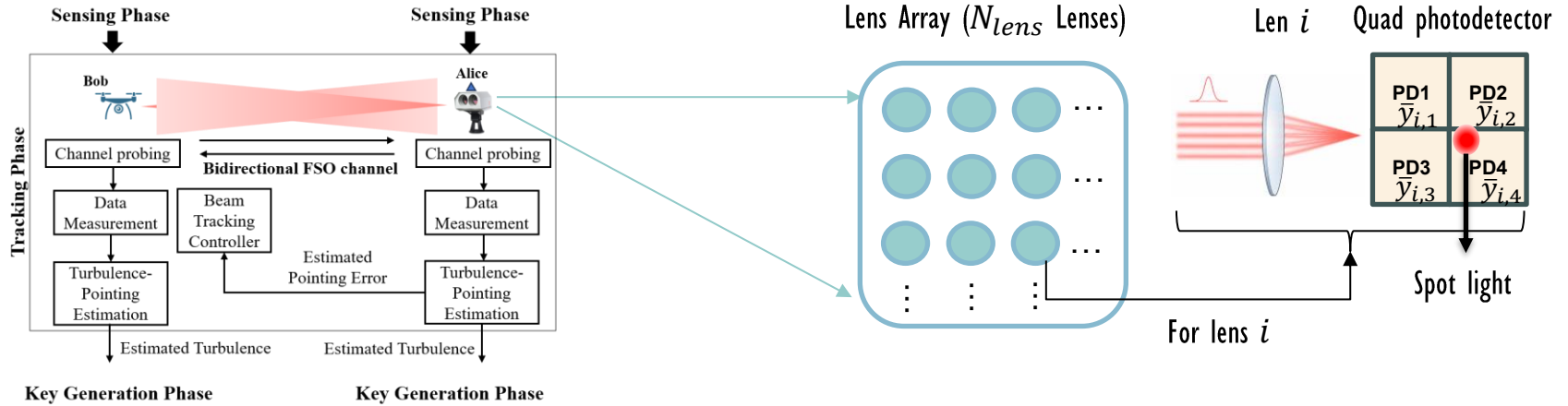
$$h_{p,i}(\theta_{ex}, \theta_{ey}) = \frac{r_a^2}{w_z^2} \exp\left(-\frac{(x_i - Z_L \theta_{ex})^2 + (y_i - Z_L \theta_{ey})^2}{w_z^2}\right)$$

- r_a : Aperture radius of each lens - Z_L : Alice and Bob distance

- w_z : Beam waist at distance Z_L - $p_i = (x_i, y_i)$: Center of lens i

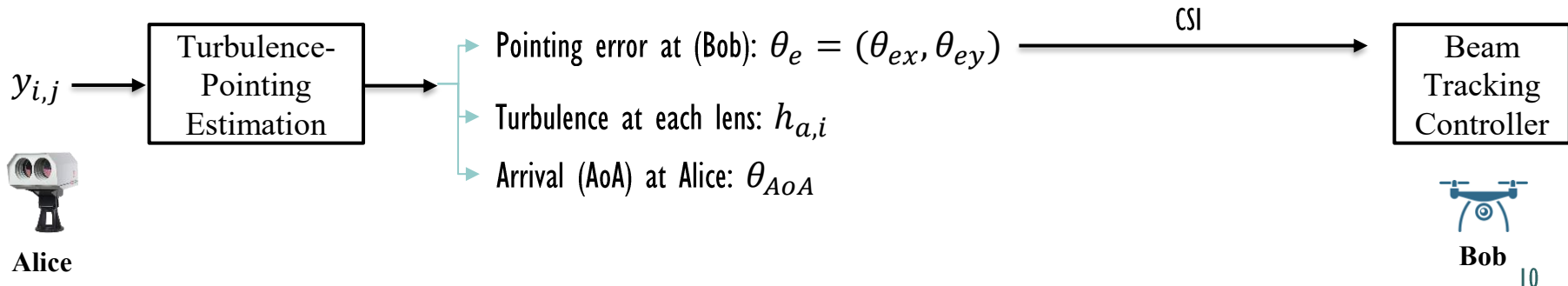
System model

2. Tracking phase:



- Channel probing and Data Measurement: Observe and measure channel
- Turbulence - Pointing Estimation: Turbulence and pointing are estimated for key generation and beam tracking
- Beam Tracking Controller adjusts Bob's pointing direction based on the estimated pointing error from Alice

The channel at PD j behind lens i is: $y_{i,j} = P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y}) h_{a,i} + n_{i,j}$



System model

2. Tracking phase: — Turbulence - Pointing Estimation

Stage I: Noise reduction

- The channel at PD j behind lens i is:

Stage II: AoA estimation

Stage III: Pointing error estimation

Stage IV: Jitter + Turbulence estimation

Pointing error at (Bob): $\theta_e = (\theta_{ex}, \theta_{ey})$

Arrival (AoA) at Alice: θ_{AoA}

Noise at the receiver

$$y_{i,j} = P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y}) h_{a,i} + n_{i,j}$$

Turbulence at each lens: $h_{a,i}$

Stage I: Average L_b samples within each channel coherence time.

$$\bar{y}_{i,j} = \frac{1}{L_b} \sum_{l=1}^{L_b} y_{i,j}[l]$$

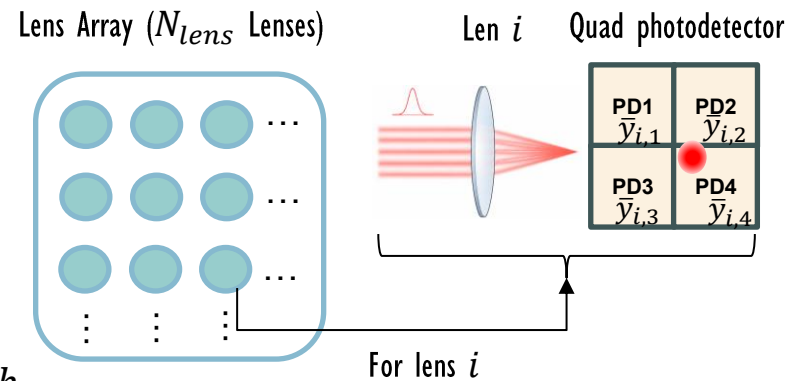
→ Averaging reduces the noise to: $\bar{n}_{i,j} \sim \mathcal{N}(0, \frac{\sigma_n^2}{L_b})$

→ The effective noise variance is reduced by $1/L_b$

After Stage I: The received signal can be approximated as noise-free.

$$\bar{y}_{i,j} \approx P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y}) h_{a,i}$$

→ Input step II: $\bar{y}_{i,j}$



System model

2. Tracking phase: — Turbulence- Pointing Estimation

Stage II: AoA estimation ($\theta_{AoA,x}, \theta_{AoA,y}$)

- The signal at PD j behind lens i is

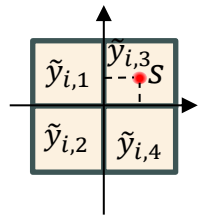
$$\bar{y}_{i,j} = P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y}) h_{a,i}$$

Step I: Removes common scaling (transmit power, pointing loss, turbulence)

$$\tilde{y}_{i,j} = \frac{\bar{y}_{i,j}}{\sum_{k=1}^4 \bar{y}_{i,k}} = \frac{P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{a,i} h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y})}{P_t \cdot h_0 \cdot h_{p,i}(\theta_{ex}, \theta_{ey}) h_{a,i} \sum_{k=1}^4 h_{AoA,i,k}(\theta_{AoA,x}, \theta_{AoA,y})} = \frac{h_{AoA,i,j}(\theta_{AoA,x}, \theta_{AoA,y})}{\sum_{k=1}^4 h_{AoA,i,k}(\theta_{AoA,x}, \theta_{AoA,y})}$$

(Pointing loss and turbulence of each PD are common)

Step II: AoA estimation



$$S = (s_x, s_y) = (f_c \cdot \theta_{AoA,x}; f_c \cdot \theta_{AoA,y})$$

S : Location of spot light on Quad PD

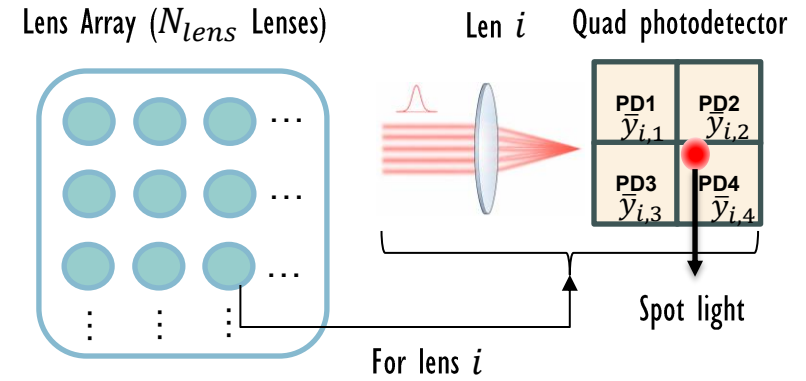
f_c : Focal length

Key idea: Based on the difference of power in each PD

→ MLP will learn to know the location of S → Estimate ($\theta_{AoA,x}, \theta_{AoA,y}$)

Step III: AoA Compensation (prepare Stage 3 input)

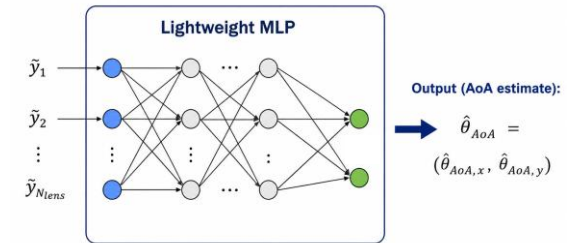
$$Z_i = h_{p,i}(\theta_{ex}, \theta_{ey}) h_{a,i}$$



25 %	25 %
25 %	25 %

20 %	30 %
20 %	30 %

40 %	25 %
25 %	10 %



Input: $\{\tilde{y}_i\}_{i=1}^{N_{lens}}$ where $\tilde{y}_i = [\tilde{y}_{i,1}; \tilde{y}_{i,2}; \tilde{y}_{i,3}; \tilde{y}_{i,4}]$

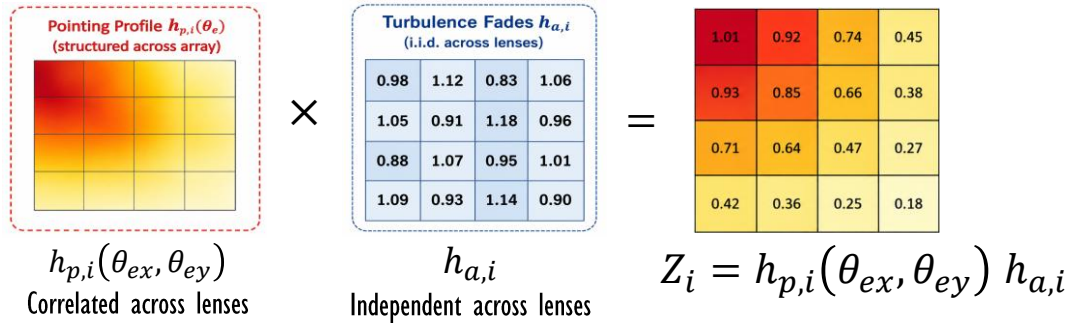
Output: AoA estimation

System model

2. Tracking phase: — Turbulence - Pointing Estimation

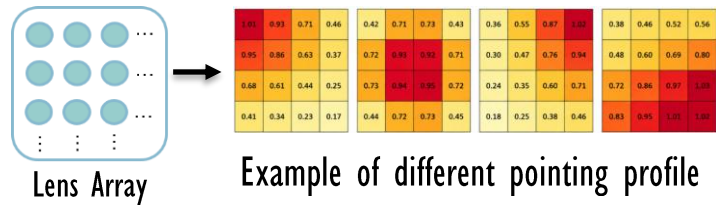
Stage III: Pointing error estimation

- The signal at lens is: $Z_i = h_{p,i}(\theta_{ex}, \theta_{ey}) h_{a,i}$



Key idea:

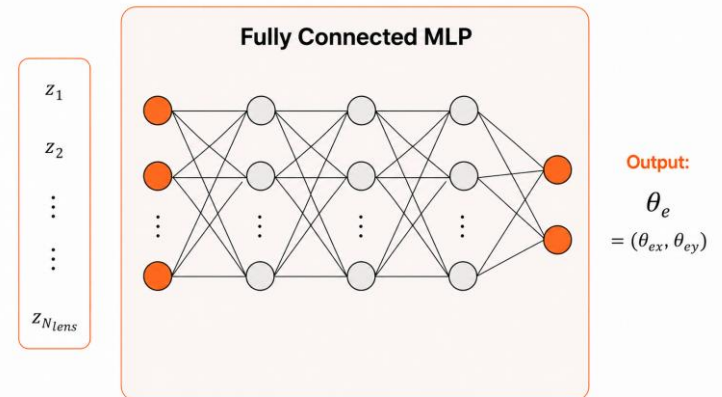
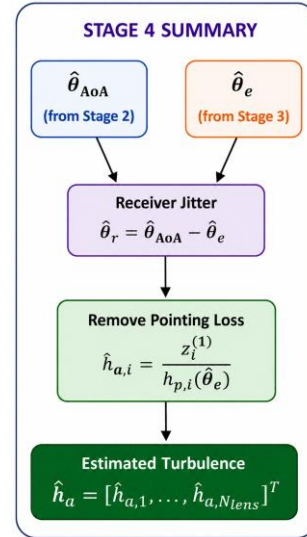
- The pointing gain is correlated across the lens array due to the Gaussian beam profile



- Turbulence is independent across lenses
 - Although turbulence distorts the signal, the spatial pointing pattern remains
- The MLP exploits this spatial pattern to estimate the pointing error $(\theta_{ex}, \theta_{ey})$

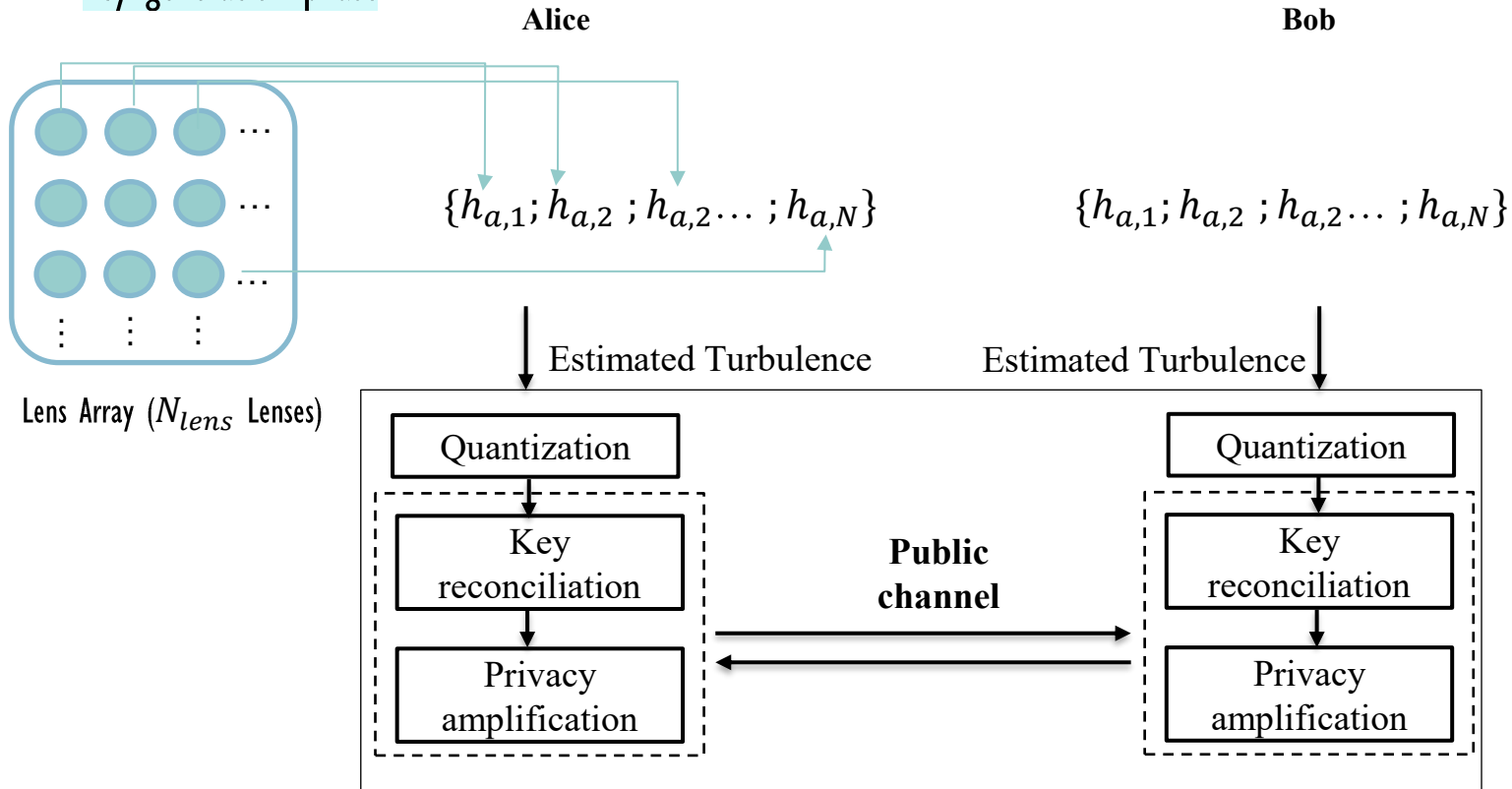
Stage IV: Turbulence estimation

- After estimating the pointing error, the pointing gain is removed from the composite channel.
- The remaining component is regarded as the turbulence channel.



System model

III. Key generation phase



- In this phase, estimated turbulence from each lens is used to create key
- The SKR is expected increase as the increase of number of lens

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Simulation results and Conclusions

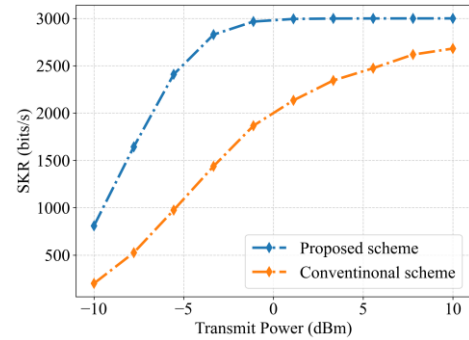


Fig. 1. SKR comparison between the conventional and proposed schemes.

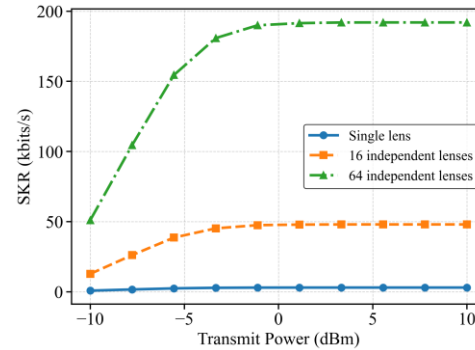


Fig. 2. SKR performance with single and multiple lenses

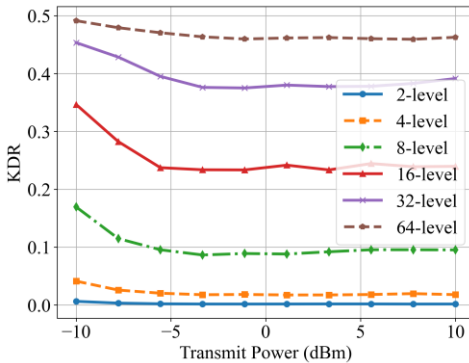


Fig. 3. KDR performance for different quantization levels.

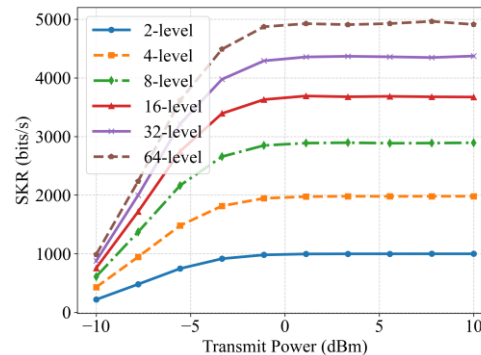


Fig. 4. SKR performance for different quantization levels.

Conventional scheme: Secret keys are generated from the received signal (pointing + turbulence).

Proposed scheme: Secret keys are generated from the estimated turbulence only.

Key insight:

Fig 1. The proposed scheme achieves a higher SKR than the conventional scheme due to enhanced key extraction.

Fig 2. Increasing the number of lenses significantly improves the SKR.

Fig 3. A higher quantization level increases the Key Disagreement Ratio (KDR).

Fig 4. A higher quantization level also improves the SKR.

Conclusion:

- Proposed a sensing-assisted SKG framework for UAV-based FSO systems
- Enabled beam tracking through multi-stage channel estimation
- Generated secret keys from the estimated turbulence instead of the received signal.
- Increased SKR by exploiting multiple lenses